Thermodynamic work capacity of quantum information processing

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Why study the thermodynamics of channels?

What is the thermodynamic cost of implementing a quantum channel?

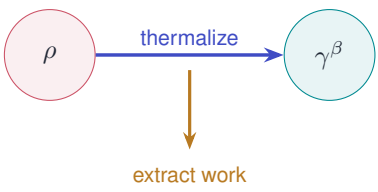
- Every component of quantum information processing is a quantum channel



- Circuit correction: $\mathcal{N} \to \mathrm{id}_m$ under thermodynamic constraints.
  manipulating a quantum circuit requires another quantum circuit.

Free energy and extractable work

Bath inverse temperature is β, and the Gibbs state is $\gamma^\beta = e^{-\beta H}/Z$.



$$F_T^\beta(\rho) = \mathrm{tr}(H\rho) - \beta^{-1}S(\rho)$$

$$\begin{aligned} F^\beta(\rho) &= F_T^\beta(\rho) - F_T^\beta(\gamma^\beta) \\ &= \beta^{-1}D(\rho\|\gamma^\beta). \end{aligned}$$

$$W_{\mathrm{ext}}^{\beta,\mathrm{rev}}(\rho \to \gamma^\beta) = F^\beta(\rho)$$

Free energy measures the thermodynamic utility of a quantum state.

How to characterize the thermodynamic utility of a quantum channel?

Thermodynamic capacity and distillation bound

Faist, Berta and Brandão quantify a process's largest thermodynamic free-energy increase:

$$C_{\mathrm{th}}(\mathcal{N}) = \max_{\rho_{A'}} \left[F^{\beta}_{T,A}\left(\mathcal{N}(\rho_{A'})\right) - F^{\beta}_{T,A'}(\rho_{A'}) \right].$$

Under Gibbs-preserving operations, one can attain this work-cost rate asymptotically.

Regula and Takagi provide universal converse bounds for distillation of unitaries from noisy channels:

$$\varepsilon \geq 1 - \mathcal{F}_{\mathbb{O}}(\mathcal{U})\mathcal{R}_{\mathbb{O}}(\mathcal{N})$$

$\mathcal{F}_{\mathbb{O}}$ is Choi-state fidelity, and $\mathcal{R}_{\mathbb{O}}$ is the robustness

Faist, Berta & Brandão, PRL 122, 200601 (2019), Regula & Takagi, Nature Communications, 12, 4411 (2021)

This work

What is a true thermodynamic resource?

In a communication setting:

it is the *preservability* of the thermodynamic resource in a presence of a reference.

* Given ρ_1 and ρ_2 with equal probability, the minimum error probability is:

$$p_{\mathrm{E}} = \frac{1}{2}\Big(\operatorname{tr}(\rho_1 \Pi_2) + \operatorname{tr}(\rho_2 \Pi_1) \Big) \leq \frac{1}{2} - \frac{1}{4} \|\rho_1 - \rho_2\|_1$$

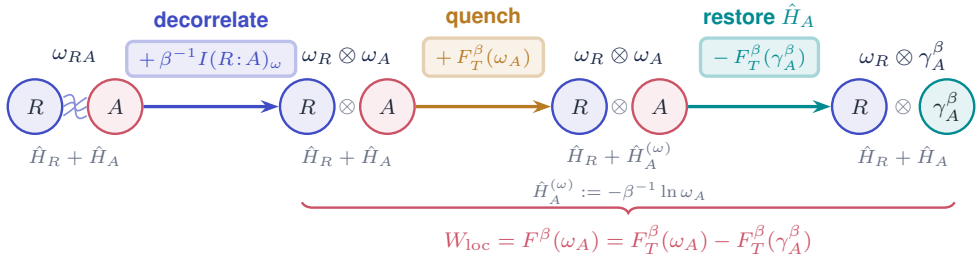
* Given two channels $\mathcal{N}_{A' \to A}$ and $\mathcal{M}_{A' \to A}$, if we access to entangled state $\rho \in RA'$, then

$$p_{\mathrm{E}} \leq \frac{1}{2} - \frac{1}{4} \sup_{\rho \in RA'} \|\mathrm{id}_R \otimes \mathcal{N}(\rho) - \mathrm{id}_R \otimes \mathcal{M}(\rho)\|_1 = \frac{1}{2} - \frac{1}{4} \|\mathcal{N} - \mathcal{M}\|_\diamond$$

True distinguishability of channels requires extensions to the reference.

Extracting work from a channel output

$$\mathrm{id}_R \otimes \mathcal{N}_{A' \to A}(\psi_{RA'}) = \omega_{RA}$$



Extractable work = channel free energy

The relative-entropy chain rule:

$$I(R\!:\!A)_\omega + D(\omega_A \| \gamma_A^\beta) = D(\omega_{RA} \| \omega_R \otimes \gamma_A^\beta).$$

Optimizing to get the work capacity of the channel:

$$\begin{aligned} W_{\text{ext}}^\beta[\mathcal{N}] &= \beta^{-1} \sup_{\psi_{RA'}} D\big((\mathrm{id}_R \otimes \mathcal{N})(\psi) \,\big\|\, (\mathrm{id}_R \otimes \mathcal{T}^\beta)(\psi)\big) \\ &= \beta^{-1} D[\mathcal{N} \| \mathcal{T}^\beta]. \end{aligned}$$

$$F^\beta[\mathcal{N}] := \beta^{-1} D[\mathcal{N} \| \mathcal{T}^\beta] = W_{\text{ext}}^\beta[\mathcal{N}]$$

continuous, additive, convex.

Resource theory of channel athermality

We consider finite-dimensional square quantum channels (for simplicity). We fix the inverse temperature of bath to $\beta \in (0, \infty)$.

* Free object: Absolutely thermal channel $\mathcal{T}^{\beta}_{A' \to A}$.
* Free operations: Gibbs preserving superchannels (GPSCs)

$$\Theta(\mathcal{T}^{\beta}_{A' \to A}) = \mathcal{T}^{\beta}_{B' \to B}.$$

* Resource monotone: $F^{\beta}[\mathcal{N}]$
* Standard resource units: For a given $\widehat{H}_A$, all unitary channels $\mathcal{U}_{A' \to A}$ have the same, highest, free energy among all channels. We consider $(\mathrm{id}_m, \mathcal{R}^{\pi})$ to be the golden unit; $\mathcal{T}^{\beta} = \mathcal{R}^{\pi}$ for $\widehat{H} \propto \mathbb{I}$.

Resource objects: Athermal channels $(\mathcal{N}_{A' \to A}, \mathcal{T}^{\beta}_{A' \to A})$.

Athermality distillation and formation cost



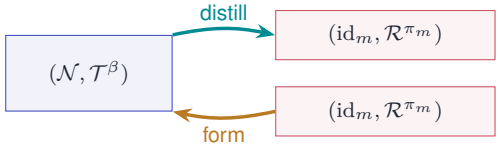
Athermality distillation and formation cost

$$\mathrm{Dist}^{\varepsilon}(\mathcal{N}, \mathcal{T}^{\beta}) := \sup_{m,\Theta} \left\{ \ln m : \; P[\Theta(\mathcal{N}), \mathrm{id}_m] \leq \varepsilon, \; \Theta(\mathcal{T}^{\beta}) = \mathcal{R}^{\pi m} \right\}.$$



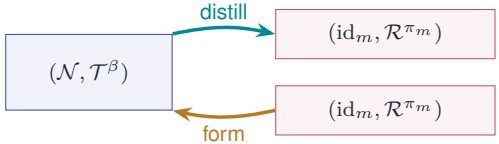
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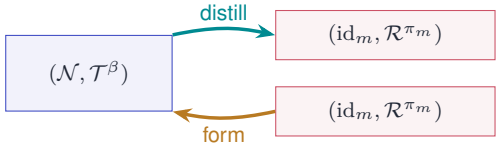
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$$\mathrm{Cost}^{\varepsilon}(\mathcal{N}, \mathcal{T}^{\beta}) := \inf_{m,\Theta} \left\{ \ln m : \; P[\Theta(\mathrm{id}_m), \mathcal{N}] \leq \varepsilon, \; \Theta(\mathcal{R}_{\pi_m}) = \mathcal{T}^{\beta} \right\}.$$

Athermality distillation and formation cost

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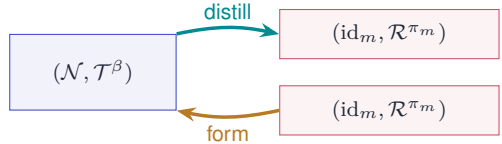
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$$\mathrm{Dist}^{\varepsilon}(\mathcal{N},\mathcal{T}^{\beta}) = \frac{1}{2}D_H^{\varepsilon^2}[\mathcal{N}\|\mathcal{T}^{\beta}]$$

$$\mathrm{Cost}^{\varepsilon}(\mathcal{N},\mathcal{T}^{\beta}) = \frac{1}{2}D_{\max}^{\varepsilon}[\mathcal{N}\|\mathcal{T}^{\beta}]$$

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$$D_H^{\varepsilon}(\rho\|\sigma) = \inf_{0\leq\Lambda\leq I}\ln\left\{\mathrm{tr}(\Lambda\sigma):\mathrm{tr}(\Lambda\rho)\geq 1-\varepsilon\right\},$$

$$D_{\max}^{\varepsilon}[\mathcal{N}\|\mathcal{M}] = \inf_{\widetilde{\mathcal{N}}\in\mathcal{B}_P^{\varepsilon}[\mathcal{N}]}\ln\left\{\lambda\geq 0:\Phi^{\widetilde{\mathcal{N}}}\leq\lambda\Phi^{\mathcal{M}}\right\}.$$

$$D_H^{\varepsilon}[\mathcal{N}\|\mathcal{M}] = \sup_{\psi_{RA'}} D_H^{\varepsilon}((\mathrm{id}_R\otimes\mathcal{N})(\psi_{RA'})\,\|\,(\mathrm{id}_R\otimes\mathcal{M})(\psi_{RA'}))\,.$$

Asymptotic reversibility

Single-shot yield and cost can differ. For many copies at fixed error $0 < \varepsilon < 1$, their leading-order rates agree:

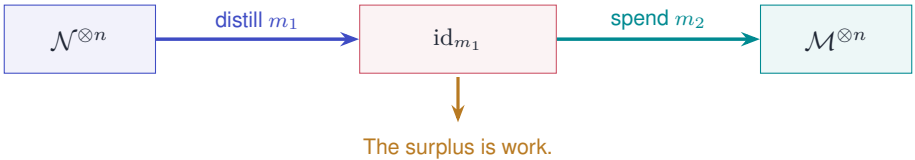
$$\limsup_{n\to\infty} \frac{1}{n}\mathrm{Dist}^{\varepsilon}(\mathcal{N}^{\otimes n}, (\mathcal{T}^{\beta})^{\otimes n}) = \frac{1}{2}D[\mathcal{N}\|\mathcal{T}^{\beta}],$$

$$\liminf_{n\to\infty} \frac{1}{n}\mathrm{Cost}^{\varepsilon}(\mathcal{N}^{\otimes n}, (\mathcal{T}^{\beta})^{\otimes n}) = \frac{1}{2}D[\mathcal{N}\|\mathcal{T}^{\beta}].$$

$$\mathrm{Dist}(\mathcal{N}, \mathcal{T}^{\beta}) = \mathrm{Cost}(\mathcal{N}, \mathcal{T}^{\beta}) = \frac{\beta}{2}F^{\beta}[\mathcal{N}]$$

Athermality conversion is reversible under Gibbs-preserving superchannels.

Channel conversion through standard units



$$W^{\varepsilon_1,\varepsilon_2}_{\beta,\mathrm{ext}}[\mathcal{N}\to\mathcal{M}] = \frac{2}{\beta}\left[\mathrm{Dist}^{\varepsilon_1}(\mathcal{N},\mathcal{T}_A^{\beta}) - \mathrm{Cost}^{\varepsilon_2}(\mathcal{M},\mathcal{T}_B^{\beta})\right].$$

The conversion work is a free-energy difference

In the asymptotic limit, the optimal net work per conversion is the difference between the two channel free energies.

$$\Delta W_{\mathrm{ext}}^{\beta}[\mathcal{N} \to \mathcal{M}] = F^{\beta}[\mathcal{N}] - F^{\beta}[\mathcal{M}]$$
$$= \beta^{-1}\left(D[\mathcal{N}\|\mathcal{T}_A^{\beta}] - D[\mathcal{M}\|\mathcal{T}_B^{\beta}]\right)$$

From states to channels: the equilibrium dictionary

Concept	States	Channels
Equilibrium	γ^β	$\mathcal{T}^\beta(X) = \mathrm{tr}(X)\gamma^\beta$
Free energy	$\beta^{-1}D(\rho\|\gamma^\beta)$	$\beta^{-1}D[\mathcal{N}\|\mathcal{T}^\beta]$
Work extraction	$\rho \to \gamma^\beta$	$\mathcal{N} \to \mathcal{T}^\beta$.
Free maps	$\mathcal{M}(\gamma^\beta) = \gamma^\beta$	$\Theta(\mathcal{T}^\beta) = \mathcal{T}^\beta$
Conversion work	$F^\beta(\rho) - F^\beta(\sigma)$	$F^\beta[\mathcal{N}] - F^\beta[\mathcal{M}]$

Where do we go from here?

- Athermality of multipartite quantum channel $\mathcal{N}_{A'B'\to AB}$, with local thermalization:

$$F[A|B]_{\mathcal{N}} = \beta^{-1} \inf_{\mathcal{Q}} D[\mathcal{N} \| \mathcal{T}^{\beta} \otimes \mathcal{Q}_{B'\to B}].$$

  Signaling properties $A' \to B$ play important role in manipulation cost.

- A stricter implementation model may change reversibility.
- Higher-order process extension: correcting a noisy multi-time quantum circuit, with a bounded energy budget.

Classical and coherent sectors of channel athermality

Known structure for states

Thermal channels reversibly interconvert energy-incoherent states in the i.i.d. limit.



$$D(\rho^P \| \gamma^\beta) = D(\sigma_X \| \pi_X)$$

State athermality = non-uniformity + coherence

Channel analogue

For for time-covariant processing, asymptotic yield and cost equal $\frac{1}{2} D[\mathcal{N} \| \mathcal{T}^\beta]$.



$$D[\mathcal{N}^P \| \mathcal{T}^\beta] = D[\mathcal{P}_X \| \mathcal{R}^{\pi_X}]$$

Athermality of classical channels is the dynamical non-uniformity.

Coherence processing ability seems to be crucial component of channel athermality

Thank you.

States: Horodecki–Oppenheim; Brandão et al.; Gour (2022). Channels: HB, Dhanuja G.S., in preparation

Resource conversion distance

How close can we go from the resource $(\mathcal{N}, \mathcal{T}_1^\beta)$ to the resource $(\mathcal{M}, \mathcal{T}_2^\beta)$, using only free operations?

Channel conversion distance:

$$d_{\mathrm{GPSC}}((\mathcal{N}, \mathcal{T}_1^\beta) \to (\mathcal{M}, \mathcal{T}_2^\beta)) = \min_{\Theta} \left\{ P[\mathcal{M}, \Theta(\mathcal{N})] \ : \ \Theta(\mathcal{T}_1^\beta) = \mathcal{T}_2^\beta \right\}.$$

where $P[\mathcal{M}, \mathcal{K}]$ is the purified channel distance between the channels $\mathcal{M}$ and $\mathcal{K}$ given by

$$P[\mathcal{M}, \mathcal{K}] := \sup_{\psi \in (RA')} P(\mathcal{M}(\psi_{RA'}), \mathcal{K}(\psi_{RA'}))$$

$P(\rho, \sigma) := \sqrt{1 - F(\rho, \sigma)}$ is purified distance.

Trace distance: $T(\rho, \sigma) := \frac{1}{2} \|\rho - \sigma\|_1$, and Fidelity: $F(\rho, \sigma) := \left\| \sqrt{\rho}\sqrt{\sigma} \right\|_1^2$.

$$1 - F(\rho, \sigma) \ \leq \ T(\rho, \sigma) \ \leq \ \sqrt{1 - F(\rho, \sigma)}.$$

If $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\phi\rangle\langle\phi|$, then $F(\rho, \sigma) = |\langle\psi|\phi\rangle|^2$.

Athermality distillation

For an error $\varepsilon \in [0,1]$ and a resource quantum channel $(\mathcal{N}, \mathcal{T}^\beta)$, the one-shot athermality distillation yield $\mathrm{Dist}^\varepsilon(\mathcal{N}, \mathcal{T}^\beta)$ under GPSCs are

$$\begin{aligned}\mathrm{Dist}^\varepsilon(\mathcal{N}, \mathcal{T}^\beta) &:= \sup_m \left\{ \ln m : \; d_{\mathrm{GPSC}}((\mathcal{N}, \mathcal{T}^\beta) \to (\mathrm{id}_m, \mathcal{R}^\pi)) \leq \varepsilon \right\} \\ &= \frac{1}{2} D_H^{\varepsilon^2}[\mathcal{N} \| \mathcal{T}^\beta],\end{aligned}$$

Use $\mathcal{R}^\pi = \frac{1}{m^2}\mathrm{id}_m + \frac{1}{m^2}\sum_{i=1}^{m^2-1}\mathcal{W}^i = \frac{1}{m^2}(\mathrm{id}_m + \mathrm{id}_m^\perp)$. A distillation channel can be written as

$$\Theta_\psi^\Lambda(\mathcal{N}) = \mathrm{tr}(\mathcal{N}(\psi)\Lambda)\mathrm{id} + \mathrm{tr}(\mathcal{N}(\psi)(\mathbb{1}_A - \Lambda))\frac{1}{m^2-1}\mathrm{id}^\perp.$$

the conversion distance is given by $d_{\Theta_\psi^\Lambda}((\mathcal{N}, \mathcal{T}^\beta) \to (\mathrm{id}, \mathcal{R}^\pi)) = \sqrt{1 - \mathrm{tr}(\mathcal{N}(\psi)\Lambda)}$

Athermality distillation

For an error $\varepsilon \in [0,1]$ and a resource quantum channel $(\mathcal{N}, \mathcal{T}^\beta)$, the one-shot athermality formation cost is given by $\mathrm{Cost}^\varepsilon(\mathcal{N}, \mathcal{T}^\beta)$ under GPSCs are

$$\begin{aligned}\mathrm{Cost}^\varepsilon(\mathcal{N}, \mathcal{T}^\beta) &:= \inf_m \left\{ \ln m : \; d_{\mathrm{GPSC}}((\mathrm{id}_m, \mathcal{R}^\pi) \to (\mathcal{N}, \mathcal{T}^\beta)) \leq \varepsilon \right\} \\ &= \frac{1}{2} D_{\max}^\varepsilon [\mathcal{N} \| \mathcal{T}^\beta].\end{aligned}$$

$\mathcal{T}^\beta = \Theta(\mathcal{R}^\pi) = \frac{1}{m^2}\Theta(\mathrm{id}_m) + \frac{1}{m^2}\Theta(\mathrm{id}_m^\perp)$. Therefore,

$$\begin{aligned}d_{\mathrm{GP}}((\mathrm{id}_m \mathcal{R}^\pi) \to (\mathcal{N}, \mathcal{T}^\beta)) &= \min_{\mathcal{E}} \left\{ P[\mathcal{N}, \mathcal{E}] \; : \; m^2 \mathcal{T}^\beta \geq \mathcal{E} \right\} \\ &= \min_{\mathcal{E}} \left\{ P[\mathcal{N}, \mathcal{E}] \; : \; 2 \ln m \geq D_{\max}[\mathcal{E} \| \mathcal{T}^\beta] \right\}\end{aligned}$$